

MTH 1400 Precalculus — Quiz 6

Time: 15 minutes Name: _____

Instructions. Show your work clearly. Give exact answers.

1. Use log properties to find the **exact value**.

$$\log_{30}(10) + \log_{30}(3)$$

$$\log_{30}(10 \cdot 3) =$$

$$\log_{30}(30) = y$$

$$30^y = 30 \Rightarrow y = 1$$

(5)

2. Use log properties to find the **exact value**.

$$\log_4(21) \cdot \log_{21}(64)$$

$$= \log_4(64) = y$$

$$\text{so } 4^y = 64$$

$$\Rightarrow y = 3$$

formula: $\log_a b \cdot \log_b c$
 $= \log_a c$

(5)

3. Let $\log_b A = 3$ and $\log_b B = -4$. Find $\log_b(AB)$.

$$\log_b AB = \log_b A + \log_b B = 3 + (-4) = -1$$

(5)

4. Write as the sum and/or difference of logarithms.

$$\log_7 \left(\frac{5}{9} \right)$$

$$= \log_7 5 - \log_7 9$$

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